



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTPP/ELECT MAINT/2023-24/EMD/LAB /751560

Date:

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Subject:REVISION OF LMI/OGN/OPS/ELEC/040 GUIDELINES ON
NUMERICAL RELAY MAINTENANCE.

**Subject: REVISION OF LMI/OGN/OPS/ELEC/040 GUIDELINES ON NUMERICAL RELAY
MAINTENANCE.**

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

Changes envisaged by COS.

May please approve.

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NTPC Limited

SIMHADRI SUPER THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/ELEC/040
ISSUE NO: 1, REVISION: 1
DATE: Dec 2023

**GUIDELINES ON NUMERICAL RELAY
MAINTENANCE**

APPROVED FOR

IMPLEMENTATION BY -----

Group General Manager (Simhadri)

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NTPC LIMITED

SIMHADRI THERMAL POWER STATION

OPERATION AND MAINTENANCE DEPARTMENT

LMI(Name):	Guidelines on Numerical Relay Maintenance		
LMI No.	LMI/OGN/OPS/ELEC/040		
ISSUE NO.	1	Revision: 1	Date: Dec 2023

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GUIDELINES ON NUMERICAL RELAY MAINTENANCE

1.0 INTRODUCTION

Protective relaying is an integral part of any electrical power system. The fundamental objective of system protection is to quickly isolate a problem so that the unaffected portions of a system can continue to function. Protective relays are the decision-making device in the protection scheme. They monitor circuit conditions and initiate protective action when an undesired condition is detected.

Initial phase of protective relays were electromechanical relays, then it evolved with static/processor-based relays and subsequently numerical relays.

1.1 PURPOSE OF PROTECTIVE RELAYING:

The primary purpose for installing protective relaying is to provide fast and selective clearing of faulted equipment or lines connected to the power system so that damage is kept to a minimum. Relays are also designed to provide secure and dependable operations so that the least amount of equipment is removed from the system when a fault occurs.

There are several reasons to use protective relaying:

- To provide alarms when measured process limits are exceeded, thereby allowing operators an opportunity to intervene with corrective actions
- To isolate faulted circuits or equipment from the remainder of the system so that the system can continue to function
- To limit damage to faulted equipment
- To minimize the possibility of fire or catastrophic damage to adjacent equipment
- To minimize hazards to personnel
- To provide post-fault information to help analyze the root cause

1.2 RELIABILITY OF PROTECTIVE RELAYS:

Under normal power system operation, protective relays remain idle and serve no active function. However, when required to operate because of a faulted or undesirable condition, it is imperative that the relays function correctly. The failure of a protective relay to contain and isolate an electrical problem can have severe plant-wide repercussions.

Another point of concern is the undesired operation of a protective relay during normal plant conditions or tolerable transients. Inadvertent relay operation can result in unnecessary system or plant downtime.

A maintenance and surveillance program will help to ensure that the protective relays respond properly to normal and abnormal conditions.

1.3 NUMERICAL RELAY ADVANTAGE:

Numerical relay has many advantages over electromechanical and static relays of which some are mentioned below.

- a) Numerical relays have multiple protection elements in one device thereby leading to reduced number of relays in a protection scheme. This means that several relays are no longer required for the various protection schemes.

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- b) Choice of alternative CT Ratings / Aux. Voltages and Characteristics in Single Execution.
- c) Enhanced Protection Functions and Related Features along with adaptive protections.
- d) Data Acquisition Capabilities (Event logging & Disturbance recording) - numerical relays store fault and sequence-of-event operation information within the device after a trip occurs. When a trip does occur, fault information must be accessible in a short period of time to minimize the outage time. Stored information typically includes RMS and instantaneous waveform capture, Vectors diagrams, Harmonics, relay protection element operation times etc. The relays have a capability to trigger the fault recording even for the pre-fault conditions.
- e) Facility of communication with other devices.
- f) Continuous self-monitoring (Internal failures are alarmed enabling quick corrective action)
- g) User Programmable Trip Logic and output Relay Configuration for internal protection functions.

However auxiliary relays cannot be eliminated. The only limiting factor is the Input / Output capacity of the relays.

2.0 SUPERCEDED DOCUMENT

LMI/OGN/OPS/ELEC/040 ISSUE NO: 1, REVISION: 0 DATE: Feb 2022

3.0 SAFETY INSTRUCTIONS:

All safety and general instructions laid down in the documentation supplied with the equipment shall be followed at times by the maintenance personnel. Some of the general safety guidelines applicable to personnel and equipment are as follows:

(a) Personnel:

- Appropriate clothing, gloves, safety boots etc.
- The personnel concerned must be familiar with the basic working regulations governing a given workstation: mechanical, dielectric, electric hazards etc.

(b) Handling/Testing Equipment:

- This must be in good working order, regularly maintained, properly adjusted and compliant with international/national standards.

(c) Tools in General:

- Only use tools appropriate to the type of work to be carried out.

(d) Working Area:

- Make sure the floor is safe (free from oil, blunt objects, water etc.).
- Obey the lock out /tag-out and permit to work procedures. Proper isolation of primary equipment to avoid any indirect hazard to working person.
- The testing area shall be properly demarcated and kept clear

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- Proper grounding of panels and testing equipment shall be ensured before commencing work.
- Workers shall need to diligently monitor for changes in safe work clearances and employ test before touch when working on the electrical portions
- Schematic drawings and test reports shall be available at working site.

4.0 NUMERICAL RELAY CONSTRUCTION:

A numerical relay digitally samples the incoming analog signals, converts them to a binary number, and passes them to a microprocessor chip where algorithms perform mathematics to measure the condition of the power system and to provide the protective (trip) functions. Microprocessors also perform control and logic functions.

4.1 NUMERICAL RELAY HARDWARE:

4.1.1 POWER SUPPLY:

The Power Supply card provides the various voltages necessary to drive the electronic components of the relay. Power supplies may be specified for different voltage applications—both AC and DC. An obvious problem occurs when a user applies the wrong level of voltage to the relay power supply. Settings, annunciations, and waveform captures remain saved even if supply voltage is lost to the relay. Internal batteries that are used commonly have a life span of 10 years. It is advisable to change these batteries prior to their end of life. The instruction manuals for the relays will be referred for mounting position of the batteries and replacement criteria.

4.1.2 MEASURING INPUTS:

In the measuring input card, currents and voltages from the incoming analog signals are converted to voltage at a level appropriate for internal processing. The number of inputs will vary depending on the type of relay for the application. These are then passed on to input amplifiers for filtering and amplification.

4.1.3 ANALOG-TO-DIGITAL:

The analog-to-digital subsection consists of a multiplexer, an analog-to-digital converter, and memory components to transmit digital signals to the microcomputer subsection.

4.1.4 MICROCOMPUTER/MICROPROCESSOR:

The microcomputer/microprocessor subsection processes the measured values and executes the actual protection and control functions. Functions include filtering, preparing measured quantities, continuous monitoring, control of signals for logic functions, output of control commands, recording of fault information, and management of the operating system.

4.1.5 DISPLAYS:

The displays have a visual display of information, light emitting diodes (LEDs) for displaying target and status information on the front panel of the relay, and diagnostic lights to indicate when the relay diagnostics have detected an error in the processing. One significant advantage of using digital relays over solid-state and electromechanical relays is the front panel interface. This interface provides the benefits of gaining access to operational information and the ability to manually apply

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setting changes locally at the relay device. This front panel interface is commonly called the human machine interface (HMI).

4.1.6 COMMUNICATIONS:

Digital relays are typically provided with a variety of communication options. These options may include multiple ports for communicating with the relay both locally and remotely, connection to a network or SCADA System, and connection through multiplexers or telephone lines. Options may include RS232, RS485, Ethernet, and multi-mode or single-mode fiber connections.

4.1.7 INPUT/OUTPUT (I/O):

The input/output (I/O) card provides a means for the numeric relay to communicate with the “outside world”. Binary inputs are optically isolated inputs that allow information regarding the status of external devices to be used in the logical decisions of the relay. Some relays may require that the nominal voltage of binary inputs be specified when ordering.

Binary outputs provide the means for the numeric relay to control external devices such as the tripping and closing of circuit breakers.

4.1.8 ENCLOSURE CASE:

Digital relays are supplied in a variety of physical sizes. Relays can be provided in surface or flush mount design. Terminal blocks for wires are typically connected on the rear of the case. A communication port for the local PC interface is typically on the front panel of the relay and any additional communication ports for SCADA, IIRIG B, and RTD boxes and modems are usually connected to the rear of the relay.

4.2 NUMERICAL RELAY SERVICE CONDITION:

Conditions that can affect relay operation can include the following:

- Ambient Temperature - The ambient temperature is the temperature of the air measured 30 cm (11.8 in.) from the outside surface of the enclosures or covers of the relay equipment.
- Altitude
- Humidity
- Fumes or vapors
- Dust
- Steam
- Salt air
- Transportation or storage conditions
- Extreme or sudden temperature changes
- Extreme supply voltage fluctuations
- Excessive electrical noise
- Electromagnetic radiation

The site conditions to be evaluated and corrective action to be planned based on impact of above conditions on service life & reliability of numerical relays.

4.3 NUMERICAL RELAY CHARACTERISTICS:

In general, the ideal protective relay would have the following design characteristics:

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- **Maximum reliability:** The relay will actuate only for equipment or system conditions indicative of a problem. Spurious relay actuation or actuation in response to normal or tolerable transient conditions will not occur.
- **Maximum speed:** The relay will detect equipment or system faults and perform its protective function as fast as possible while maintaining selectivity and security.
- **Selectivity:** Only the faulted equipment will be removed from service by the protective function—minimum system disruption occurs as a result of relay actuation.
- **Simplicity:** The relay protection scheme will be designed as simply as possible. It is extremely easy to over-design a protection scheme using multifunctional digital relays. Care must be taken not to have too much redundancy that could lead to unnecessary complexity of the overall system design.
- **Provide Information:** The digital relays must be able to provide the user with information regarding a fault or event. The relays have the capability to record events that include plant status, protective function tripping data, and analog data representing the currents and voltages connected to the relay. This is one of the main advantages of numerical protective relays. The more accurate and accessible the information, the easier it is to analyze and make informed decisions to return a machine to service.

4.4 TIME SYNCHRONIZATION:

Numerical relays have the option to provide time-of-day information in synchronous form for data correlation with time. This feature is important when evaluating system-wide transients. Parallel and serial formatted time codes are used to efficiently interface the timing system (time-of-day source) to the user system. Parallel time codes are defined in IRIG Standard 205-87.

With the IRIG communication capability, multiple relays can be time-synchronized by using an IRIG B clock reference. This allows for comparison between multiple relay events and fault recordings without introducing timing errors. Several other protocols are also available for time synchronization like SNTP etc. Time synchronization for future relay installation will be done preferably through networking accessed by a centralized PC and time synchronization will be done.

5.0 MAINTENANCE STRATEGY OF NUMERICAL RELAYS:

5.1 MAINTENANCE REQUIREMENT:

An effective maintenance program for protective relays accomplishes two primary goals.

- 1) It provides a high degree of confidence that the electrical power protection system will respond to abnormal conditions as designed. Periodic assurance that protective relays are in an operable status is particularly important.
- 2) An effective maintenance program preserves the relay’s readiness and helps to counteract normal and abnormal in-service deterioration that can affect a relay’s electronic components over time. Even under normal conditions, electrical, thermal, and environmental stresses are continually at work, slowly degrading the relays. This deterioration is much slower in numerical relays than in electromechanical relays because numerical relays are not affected by mechanical deterioration.

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Earlier electromechanical relays had moving mechanical parts and electrical components, concerns existed that failures or relay setting drift could occur over time. Hence the maintenance practices include frequent calibration and functional tests including visual inspection, cleaning of contacts and moving parts.

However, Numerical relays from its design have eliminated the issues of moving parts, contacts, coils, and drift problems. So, the focus of testing is more on functional or scheme testing than calibration.

A structured and consistent relay maintenance program is in set up for numerical relay schemes so that reliable protection operation is maintained.

Routine maintenance checks help to identify any of the deterioration in the device. Failed devices shall be replaced with new tested devices. Failed devices or device components such as printed circuit boards shall be sent to the supplier for root cause analysis and repair/replacement.

5.2 CLASSIFICATION OF MAINTENANCE:

Maintenance and testing of protective relays can be divided into four categories.

5.2.1 ACCEPTANCE TESTING: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

Acceptance testing is associated with the design and manufacture of products. A receipt inspection will be generally done to ensure that no shipping damage has occurred. However, it is necessary to perform complete calibration and electrical tests on all new relays receipt for verifying replacement suitability and performance characteristics for implemented site settings.

5.2.2 INSTALLATION CHECKS: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

Installation checks are conducted when a protective relay is initially placed in service. Installation checks include inspections and electrical tests. These checks shall be conducted to ensure that the proper settings and configuration parameters have been downloaded to the relay and perform its intended function.

5.2.2.1 VISUAL INSPECTION AND CLEANING:

When relays are received on site for a new installation, a visual inspection shall be done of the first tasks performed to identify conspicuous signs of unusual degradation. Visual checks include correct relay model, power supply voltage, and binary input/output voltage identification.

5.2.2.2 ELECTRICAL TESTS:

Electrical tests shall be performed to verify that performance characteristics are acceptable under a controlled set of input conditions (Binary & Analog inputs). Electrical tests confirm that the correct protection settings are applied to the relay and that the proper output contact is actuated by each protection function. Electrical testing of protective relays requires specialized test equipment.

Before conducting electrical tests, approved/actual configuration & settings shall be uploaded on numerical relays. Testing engineer shall be equipped with master drawings and relay configuration files. All inputs and outputs to be

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checked for proper functioning through testing kit. Protection alarms and trips shall be adequately tested and recorded in record formats. Records formats shall be detailed procedure which includes all types of isolation, simulation, normalization steps etc.

5.2.3 IN-SERVICE MEASUREMENT READINGS: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

One of the benefits of numerical relay is that it allows in-service readings through relay management software and HMI. The in-service readings gives valuable information which could avoid potential tripping like increased differential current due to CT secondary issues or binary input malfunction etc. A periodical visit shall be performed for numerical relays with a checklist for recording healthiness of current, voltages, binary inputs/outputs, differential currents etc. The disturbance records of relay shall also be triggered and analyzed for relay healthiness at regular intervals especially before overhauling/relay maintenance.

5.2.4 PREVENTIVE (OR PERIODIC) MAINTENANCE: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

Preventive maintenance shall be performed on a scheduled basis for the following reasons:

- To detect and correct impending problems
- To verify that relay performance is within acceptable operating limits
- To reduce the likelihood of degraded performance by ensuring relay is in good working order

Reliability centered numerical relay maintenance shall also be planned as per Annexure-II.

Preventive maintenance shall include:

5.2.4.1 VISUAL INSPECTION AND CLEANING: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

Visual checks as per 5.2.2.1 after the thorough cleaning of panels using vacuum cleaner to avoid dust gets accumulating inside numerical relays. If air blower is to be used, then proper sealing of numerical relays to be ensured before blowing. Thermograph inspection shall also be done on regular intervals to identify any impending problem.

5.2.4.2 ELECTRICAL TESTS: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

Once performed as a part of initial commissioning, other methods of verifying relay operation (such as event logs report scrutiny or go/no-go functional tests) shall be performed as per sub sections below. Any modifications to logic or firmware update may require thorough electrical testing as per 5.2.2.2 to ensure that no unintended side effects are introduced into the overall operation of the logic.

5.2.4.2.1 FUNCTIONAL CHECKS:

For protective relays, a functional check will be conducted to verify that the desired protective action occurs in response to a relay operation. Functional

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tests shall be accomplished using current and voltage injection to operate the output contacts & LED for verifying the circuit. All inputs shall be verified by either through source simulation or locally short/open the contact. These checks will ensure the healthiness of relay output contacts, correct configuration of alarms, trips, event logs and disturbance records.

5.2.4.2.2 SCHEME CHECKS:

A scheme check will be performed to verify the circuit wiring healthiness. All inputs shall be checked preferably from the field/remote device and upto the relay (all alarm or trip verifications). All the output to be verified from relay to the end terminations as per scheme. These checks will ensure the healthiness of cabling, relay contacts, correct configuration of alarms, trips, event logs and disturbance records.

5.2.4.2.3 SETTING AND CONFIGURATION CHECKS:

The correct settings and configuration are one of the major criteria for reliable operation of numerical relays. Hence concerned engineer will maintain a master database of settings and configuration of each relay. During periodic maintenance after all works are completed, engineer shall verify/compare the settings & configuration with the master database

5.2.4.3 RELAY DATA LOGS: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

The relay provides a great deal of information through operational indications, alarms, event logs, measured values and fault recording/ oscillography data. This information can be read out and is available from the relay's memory using a personal computer and relay management software.

Relay data logs are an effective tool for troubleshooting, analyzing the fault and operational behavior of the power system. Event/trip logs will be large enough to store an adequate number of events for an engineer to track the system performance over a period of time. The event log and fault recording time (pre fault and post fault) in numerical relays are programmable.

During electrical testing as per 5.2.4.2, all the relay logs/data shall be read and verify its effectiveness.

5.2.5 POST REPAIR OR REPLACEMENT CHECKS: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

Numerical relays are provided with watchdog contacts and can have self-monitoring to immediately signal a relay problem or failure. If it is determined that the relay requires repair, then same shall be taken out of service. Faulty relay shall be replaced with working and tested spare relay. The relay and that the associated primary equipment shall be isolated while the repair tasks are being performed. Post-repair or replacement, installation checks as per 5.2.2 shall be done.

5.3 DETERMINING MAINTENANCE INTERVALS: RESPONSIBILITY-AREA ENGINEER (EMD-LAB)

Microprocessor relays may use internal batteries to maintain nonvolatile memory. Manufacturers recommend changing the batteries prior to their end of life. So, a plan to be maintained for battery replacement as per manual instructions.

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The periodic or periodic maintenance interval will be maintained as per OD-OPS-ELEC-003.

Testing engineer shall maintain relay specific test schedules and protocols with clear instructions for safety and isolations/normalizations. The Annexure-1 may be used as references for preparing numerical relay maintenance and replacement procedures.

6.0 THE EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS:

Any deviation shall be reported to HOD (EM).

7.0 RECORDS: LIST OF RECORDS

SL NO	REF. NO	DESCRIPTION/TITLE
1	1802/T-5	Testing of Protective Relays of 11kV, 3.3 kV and 415V system
2	1806/T-5	Testing of Protective Relays of GRP
3	FC-1A8	Testing of Stage-II EHV Relays (400KV,33 kV), Stage-1 EHV Relays (400KV,33 kV) & Distance Relays.

8.0 REVIEW:

Documents will be reviewed once in two years by HOP (Simhadri).

9.0 RECORDS OF REVISION:

- a) Restructuring of content in new OGN format for uniformity.
- b) Annexure-I is updated for additional checks during preventive maintenance and replacement of relays. Procedure for replacement of existing electromechanical/static with numerical relays is also included.
- c)Annexure-III added for reference checklist for replacement of relay.

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10.0 CHECK LIST:

RELAY TESING CHECK LIST										
Unit:			location:					Date:		
SL N O	NAME OF RELAY	PROTECTION	PICK UP VALUE	TRIP VALUE/ TIME	ALARM	TRIP	DR	BKR STATUS	86/NON 86	REMARKS

11.0 ANNEXURE LIST:

SL. NO.	DESCRIPTION	ANNEXURE
1	NUMERICAL RELAY MAINTENANCE GUIDELINES	I
2	RELIABILITY CENTERED MAINTENANCE (RCM)	II
3	REFERENCE CHECK LIST FOR REPLACEMENT OF RELAY	III

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ANNEXURE-I

NUMERICAL RELAY MAINTENANCE GUIDELINES

A) PREVENTIVE MAINTENANCE GUIDELINES

1. Ensure availability of master database (which will be latest considering all modifications/recommendations) of relay settings, configuration matrix and logics. This database availability at station shall be the responsibility of area protection engineer/HOD.
 2. Manually Trigger Disturbance Record in numerical relays which are going under maintenance (Some relays have manual trigger option and if same is not available then use spare DI configured for DR). Check and verify the status of all binary inputs and outputs in running condition especially in GRP/Switchyard relays. Extracted DR's shall be evaluated by site engineer. DR database shall be kept for GRP, HT Switchgear Incomers and Switchyard where comparison with trip DR is frequently required.
 3. Take back up of the numerical relay along with its process data (event log/alarm log/relevant DRs). The relay configuration shall be checked with the master database for any changes.
 4. After verification if no deviations found then relay testing shall be planned as per OD-OPS-ELEC-003. The changes if found shall be investigated.
 5. A copy of Latest OS recommendations may be kept available. Deviations if any w.r.t latest corporate OS Guidelines may be rectified. Master database to be updated accordingly.
 6. Before relay testing ensure cleaning of panel done, tightness of all TB's and cable lug healthiness checked, all earthing wires are intact, relay terminations are checked for tightness.
 7. Ensure availability of testing kit, testing format (inclusion of all checkpoints to avoid any auxiliary tripping due to testing direct/indirect impact), and tools if required.
 8. Compare the operational measured values (V, I, Watts, or Vars) indicated by the relay and h the injected analogue parameters from test kit.
 9. The Current/Voltages may be injected through Test Plugs or terminal blocks. It is preferable to inject through test plugs so that only desired relays will get injection voltages/currents however, if the option of Test Plugs is not available ensure the other Protection which are interlinked to the injection are in defeated condition (like CT circuit in series with LBB relay in some schemes etc.).
 10. During relay secondary injection testing ensure tripping of master trip relays on each protection trip and not on any alarm or any protection pickup. If required other protections may be deactivated which can interfere with the protection function under test. Read out and evaluate the event log and fault recordings. If required any DCS freezing of inputs is to be done to avoid any unintended tripping/operation.
- Note: a) Create an alternate setting file to ensure that the master file does not get modified when maintenance is being done. b) Ensure all the required protections are in activated condition after completion of testing.
11. Any deviation if observed during the injection or simulation testing shall be properly documented.

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12. Binary input threshold voltage settings/debounce time in the relay may be cross verified for proper intended operation. The binary inputs and binary outputs to be checked for their desired operation as per scheme up to the end point. Ensure alarm and trip occurrence on DCS/SCADA/event logger during testing.

13. All auxiliary relays to be checked for healthiness of coil and output contacts. Check pick up voltage of master trip relays which shall be within range as per relay specification. If the output contacts of the relay are directly given for breaker tripping, ensure the O/P Contact is sufficiently rated for breaker trip coil.

14. After complete testing, the relay configuration and setting file SHALL be downloaded from relay and verified with master files. Any deviations if found shall be noted and corrected.

B) PROCEDURE FOR REPLACEMENT OF NUMERICAL RELAYS WITH NUMERICAL RELAYS UPON FAILURE.

1. It is recommended numerical relays shall not be replaced with equipment in service.
2. Ensure that the master relay setting file, protection scheme, relay test equipment, personal computer, and communication cable are available.
3. Review schematics and/or circuit diagrams as appropriate to ensure that the consequences of removing the relay from service are known. Some inputs of the relay may have been looped to other in-service devices and security of same should be ensured like PT circuit loop in HT switchgear. Ensure that CT circuit current is zero using leakage Clamp meter and it should not get open circuited during replacement. Ensure PT circuit should not get short circuited by isolating the PT Circuit during replacement.
4. Review internal relay programmable logic and communication networking to ensure that automatic control or tripping logic will not have an adverse effect when the relay is removed from service.
5. Contact any SCADA/load dispatchers who may be receiving data from the relay and notify them that the data will be lost while the relay is taken out of service. Disconnect communication from the relay to SCADA so that false data are not supplied to SCADA while the relay is in test.
6. Retrieve all the relay database including event and fault/disturbance record files that may be in the faulty relay wherever possible. Note down the faulty relay condition i.e its binary input/output and watch dog status. Check the current input and voltage input at relay.
7. Identify if the relay front panel is working. If working, then check for any internal failure or any error code. Record the data of failure.
8. Ensure that plant conditions are consistent with the work shall be performed and obtain permission to remove the relay from service. Always be aware of any abnormal plant condition or system lineup that may be impacted by removing the relay from service.
9. Isolate the relay from external circuits. This action is completed by removing the attached test plug/switches or opening panel terminal blocks. An inadvertent operation may occur if the proper sequence is not followed. Outputs capable of generating trips (trip outputs, BFI [breaker failure initiate] output, or DTT [direct transfer trip] output) shall be defeated prior to opening test switches for operating and restraining quantities. Due care must be taken not to earth or short circuit any external circuit while working on relay replacement.

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10. New relay shall be preconfigured and thoroughly tested at LAB to avoid delays during replacement. However, utmost care shall be taken while transporting from LAB to location to avoid failure due to static voltages or mechanical jerk. The engineer shall record the faulty and new relay serial number and firmware version numbers. If one to one relay replacement is planned, the correctness of existing relay model Number & the new Relay Model Number shall be ensured to avoid maloperations.

11. Confirm the supply voltage is adequate for the new relay being installed. Reconnect power supply, only inputs and CT's/PT's wires to relay. Then make Relay ON to observe status of connected inputs, watch dog, outputs. Download all the settings/logics from the new relay and compare with master database.

12. Now make relay OFF and connect all output wires. After verifying all tightness make the relay ON. Before making relay ON inform and discuss with operation for any consequence/corrective action of any undesired output ON. (If required any DCS freezing of inputs is to be done to avoid any unintended tripping).

13. After the relay is made ON, check the availability of inputs/outputs to all the destinations like DCS, SCADA, other relay on network etc. Binary input threshold voltage settings/debounce time in the relay may be cross verified for proper intended operation. (If required any DCS freezing of inputs is to be done to avoid any unintended tripping/operation).

14. Post replacement of Numerical Relay Scheme Checks to be performed to verify the circuit wiring healthiness. All the output to be verified from relay to the end terminations as per scheme. These checks will ensure the healthiness of cabling, relay contacts, correct configuration of alarms, trips, event logs and disturbance records. All inputs to be verified by either through source simulation or locally short/open the contact. These checks will ensure correctness of inputs to the relay.

15. After complete verification, the site shall be cleared from all man and material before returning the permit to work.

16. Monitor the relay performance/ behavior during any associated equipment or system disturbance for proper response. The root causes analysis of faulty relay to be done to avoid repetition of failure.

17. As built and testing reports are to be maintained for preventive maintenance and reference purposes in future.

C) PROCEDURE FOR REPLACEMENT OF EXISTING CONVENTIONAL RELAYS WITH NUMERICAL RELAYS.

1. It is recommended numerical relays shall not be replaced with equipment in service.

2. It shall be ensured that approved scheme and setting file shall be made available as per relay model before replacement plan during opportunity/overhauling. Preferably one relay shall be tested as per configuration in lab. If a new make/model relay being installed, then a reference relay file from other station where it is in service and proven may be taken for comparison and adoption.

3. Ensure that the relay setting, protection scheme, relay test equipment, personal computer, and communication cable are available.

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4. Review schematics and/or circuit diagrams as appropriate to ensure that the consequences of removing the relay from service are known. Some inputs of the relay may have been looped to other in-service devices and security of same should be ensured like PT circuit loop in HT switchgear. Ensure that CT circuit current is zero using leakage Clamp meter. Ensure PT circuit should not get short circuited by isolating the PT Circuit.

5. Ensure that plant conditions are consistent with the work to be performed and obtain permission to remove the relay from service. Always be aware of any abnormal plant condition or system lineup that may be impacted by removing the relay from service.

6. Isolate the relay from external circuits. This action is performed by Switching OFF the DC circuits, disconnecting the CT/PT terminals. Disconnect the DCS related cables to avoid earthing of DCS cables.

7. Ensure Zero Voltage/Current at the relay terminals & Note down the existing relay settings, wiring details including ferrules before removal.

8. Due care must be taken not to earth or short circuit any external circuit while working on relay replacement. Remove the old Cables & Relays from the Panel.

9. Install the new relay & complete the wiring as per the new Drawing.

10. Confirm the supply voltage is adequate for the new relay being installed. Reconnect earthing wire, power supply to observe status of connected inputs, watch dog, outputs.

11. Before making relay ON inform and discuss with operation for any consequence/corrective action of any undesired output ON. (If required any DCS freezing of inputs is to be done to avoid any unintended tripping).

12. After the relay is made ON, check the availability of inputs/outputs to all the destinations like DCS, SCADA, other relay on network etc. Binary input threshold voltage settings/debounce time in the relay may be cross verified for proper intended operation.

13. Post replacement of Numerical Relay Scheme Checks to be performed to verify the circuit wiring healthiness. All the output to be verified from relay to the end terminations as per scheme. These checks will ensure the healthiness of cabling, relay contacts, correct configuration of alarms, trips, event logs and disturbance records. All inputs to be verified by either through source simulation or locally short/open the contact. These checks will ensure correctness of inputs to the relay.

14. After complete verification, the site to be cleared from all man and material before returning the permit to work.

15. As built and testing reports are to be maintained for preventive maintenance and reference purposes in future.

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ANNEXURE-II

RELIABILITY CENTERED MAINTENANCE (RCM)

Reliability-centered maintenance evaluation will include the following:

- Failure mode and effects analysis (Some failure modes are mentioned below)
- Critical failure analysis
- Root cause analysis
- Mean time between repair
- Spare parts strategy
- Maintenance plan

The focus of the RCM programs on numerical relays shall include components of the relay along with binary input and output operation connected from the relay to the external equipment.

Part of the predictive maintenance plan will be a database that tracks each relay’s firmware and hardware version numbers to identify problems with batch manufacturing and to verify that the proper firmware is in all devices. The relay firmware changes can take place over time by manufacturer to resolve unique protection problems that were undetected in earlier version devices. Relay failure history shall be kept at station with all relevant details like model, firmware, DR, events, RCA report etc.

Relay monitoring shall be through metering and status data to a DCS system. All relay self-check contacts will be wired into a DCS system or local annunciator for immediate notification.

Correct operation logs and DR shall also be stored because these testify that the device operated as intended and provides a certain level of setting/configuration verification. For an unexpected operation, the event report will be part of the documentation for a required setting change.

FAILURE MODES OF NUMERICAL RELAYS:

1. ELECTRICAL STRESSES:

The binary output on numerical relays can be destroyed quite easily. Most output contact failures occur when attempting to break heavy currents.

The most common problem occurs where the relay contact is directly connected to the circuit breaker (CB) trip coil. There will always be CB auxiliary contacts in series with the trip coil and the relay trip contacts. The purpose of the CB auxiliary contacts is to break the electrical circuit. The CB auxiliary contacts have enough traveling distance to quench the arc created by the back EMF of the trip coil. The relay contact must remain closed until the breaker auxiliary contacts opened the circuit.

The numerical relay binary inputs/outputs and analog inputs/outputs will be configured within the product’s specified parameters for relays trouble-free operation.

2. THERMAL AND ENVIRONMENTAL STRESSES:

Relays located in plant areas with elevated ambient temperatures are subject to material degradation by thermal aging process like LCD displays frequent failures. The environmental stresses that can contribute to degradation of relays are humidity, dirt, dust, oxidation, and

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chemical contaminants. Humidity can cause material degradation (rusting) and can even contribute to short circuits or arcing if sufficient condensation occurs on live electrical paths.

The temperature inside switchgear and panels may be significantly greater than the room ambient temperature if air is not circulated. Hence the air washer units with dehumidifiers shall be maintained functional to keep the switchgear room moisture and dust free with adequate ambient temperature. All the openings of the relay panels shall be properly sealed to avoid ingress of moisture, steam, dust, rodents etc.

3. POWER SUPPLY STRESSES:

Power supply card failure of numerical relays is a major single point failure. Improper power supply input to numerical relay generally lead to power supply card failure. Hence before powering up a new or replaced numerical relay, the power supply will be measured and matched with relay power supply specifications.

Fail-safe alarm contacts and loss of SCADA communications provide immediate indication of power supply failures, allowing corrective action to be initiated quickly.

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ANNEXURE-III

REFERENCE CHECK LIST FOR REPLACEMENT OF RELAY

Sl no	Details of checks to be done	Complied/Checked (Yes/No/NA)	Remarks
1	Master/approved drawing list available		
2	Master/approved setting list available		
3	Master/approved configuration file available		
4	Risk mitigation plan done and discussed with HOD as per Annexure-1		
5	Relay test equipment available		
6	Relay testing/replacement format available		
7	Back up of existing relay (setting, DR/event logs and configuration) taken		
8	Communication to be given to SCADA users (LDC/operation) for the observation of alarms if any.		
9	Note down error codes or observation of present in-service relay condition		
10	Permit received and sufficient precautions taken for in and near the working panel		
11	The relay which is to be installed is thoroughly tested and verified for configuration in lab.		
12	Isolation from other circuits like busbar, DCS, PLC, inter-trip, SCADA as identified in risk plan is done		
13	Check the input auxiliary supply for new relay is correct as per relay model		
14	Check the binary input voltage configuration is adequate as per input voltages (24/48/110/220V)		
15	Replacement is done as per format and tested. All alarms and events checked at all interconnected systems.		
16	All man and material removed before permit cancellation		
17	Monitor relay DR, inputs, outputs and event logs after it is taken in service.		
18	As built drawing and setting kept in secured location		
19	RCA of faulty relay to be done in lab or sent to OEM. Copy of email to OEM about incidence with details of relay given to COS- EL & Engineering.		

Area Engineer (EMD)

HOD(EMD)